

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset $\text{\LaTeX}$ solutions.

4.a

A is an $m \times n$ matrix r is the rank of matrix A (linearly independent rows or columns) θ_i is the i -th singular value u_i is the i -th left singular vector from orthonormal basis of $\mathbb{R}^m$ v_i^T is the i -th right singular vector from orthonormal basis of $\mathbb{R}^n$

We can reach the equation in question by matrix multiplication. $U * D$ scales each column in U by θ_i ($\theta_1 * u_1, \theta_2 * u_2, \dots, \theta_r * u_r$) Multiplying that by V^T gives us the final result. $\theta_1 * u_1 * v_1^T + \theta_2 * u_2 * v_2^T + \dots + \theta_r * u_r * v_r^T$

4.b

We can achieve this definition by following the previous summation To isolate u_i , we can multiply both sides by v_i
 $A * v_i = \sum_{j=1}^r \theta_j * u_j * v_j^T * v_i$ Since v is an orthonormal basis, $v_j^T * v_i = 1$ if $j = i$, and 0 otherwise Therefore,
 $A * v_i = \theta_i * u_i$ Then we divide both sides by θ_i to get $u_i = (1/\theta_i) * A * v_i$

4.c

From b we can see that $\theta_i * u_i = A * v_i$. Using that in the summation, we have $A_k = \sum_{i=1}^k (A * v_i) * v_i^T$ representing the projection of rows of A onto the first k right singular vectors.

4.d

To show that A_k minimizes $\|A - B\|_F^2$ as B for all B of rank k , we can use the fact that the closest point in a subspace to a matrix is its orthogonal projection onto that subspace. The projection of A onto V_k , the best-fit subspace which minimizes error among all subspaces of dimension k , produces A_k which provides the smallest possible distance from A .